MATH 130

Circles: Quiz 8 preparation

Monday, October 19, 2026 · Angles, conversions, arcs, sectors, and speed

Today’s practice sequence

Read the circle diagram

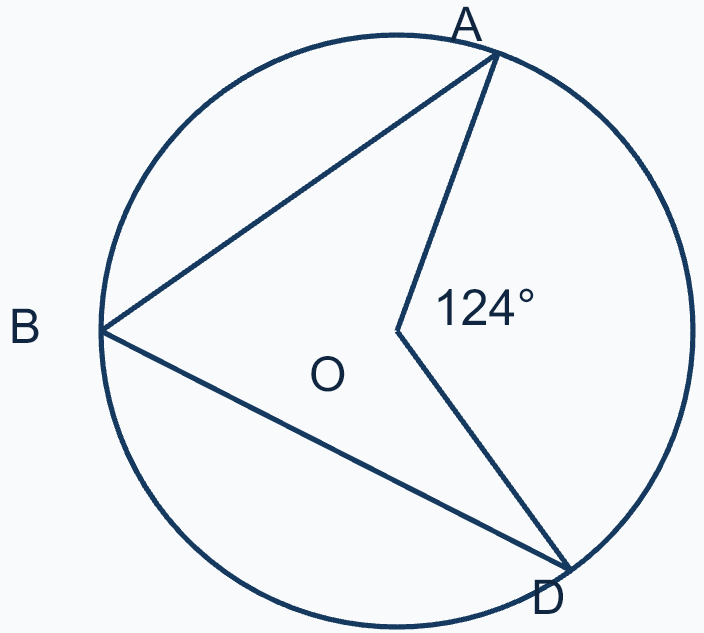
Convert the angle and keep precision

Find the unknown in an arc or sector

Convert rpm and use the radius

Where is the vertex?

Locate O and B. Which angle is central and which is inscribed?

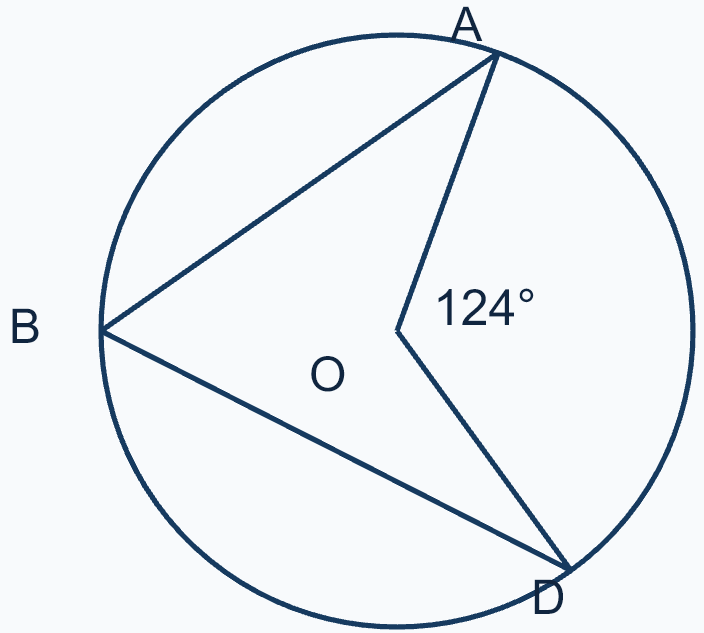


Angles and intercepted arcs

A central angle equals the measure of its intercepted arc.

An inscribed angle is half its intercepted arc.

All arcs around a circle total 360°.



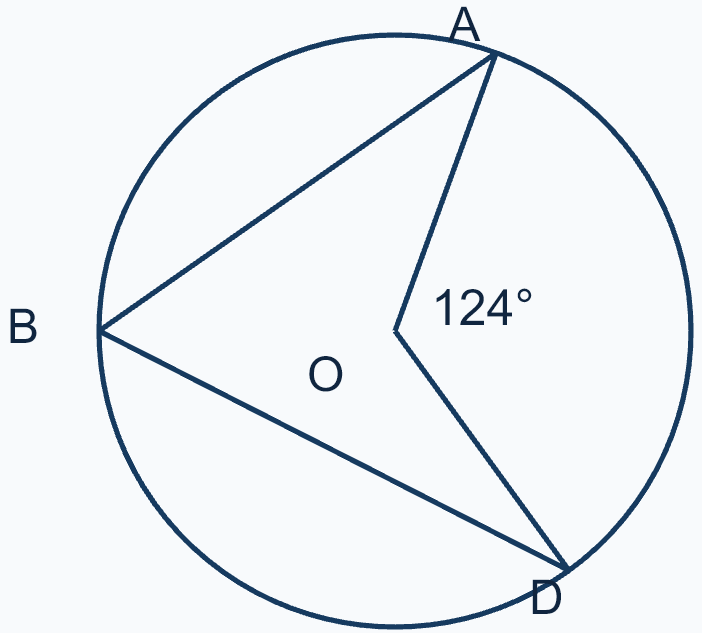
Central and inscribed angles

Find angle ABD and minor arc AD.

The central angle AOD is 124°, so minor arc AD is 124°.

Angle ABD intercepts that same arc: 124°/2=62°.

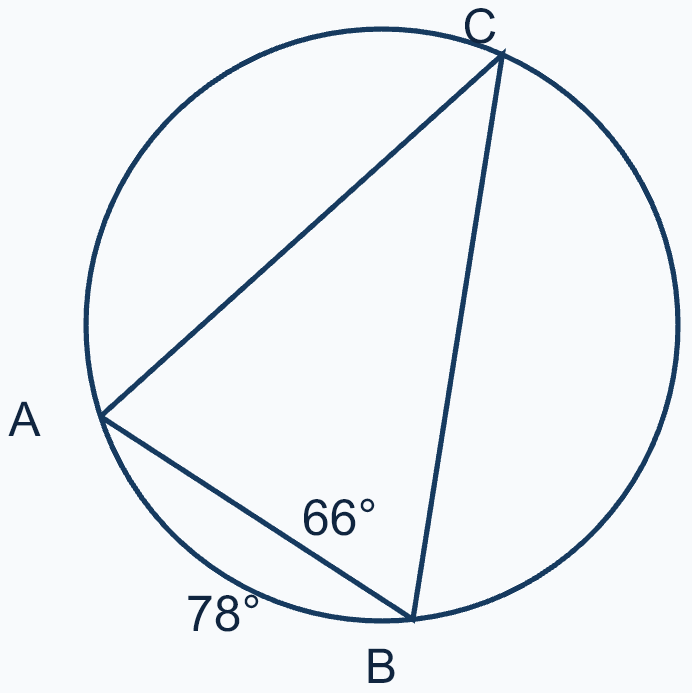
The vertex location determines the rule.



Triangle and arc practice

Find angle BAC and minor arc BC.

Show the intermediate arc or triangle-angle calculation.



Two routes to the same angle

Arc route: double the inscribed angle, subtract known arcs from 360°, then halve the remaining arc.

Triangle route: angle ACB is 39°, so angle BAC is $180° - 66° - 39° = 75°$.

Circle-angle checkpoint

Name the vertex and intercepted arc before calculating.

Central angle = arc. Inscribed angle = half the arc.

Which conversion is needed?

What changes when an angle is written as DMS, decimal degrees, or radians?

Which form is required in $s=r\theta$?

Angle conversion rules

$1^\circ = 60'$ and $1' = 60''$.

DMS to decimal: $D + M/60 + S/3600$.

Degrees to radians: multiply by $\pi/180$.

Radians to degrees: multiply by $180/\pi$.

Decimal degrees to DMS

Convert 38.726145° to DMS, to the nearest second.

Degrees: 38°. Multiply the decimal part by 60: $0.726145(60)=43.5687$.

Minutes: 43′. Multiply the remaining decimal by 60: $0.5687(60)=34.122$.

Round once, at the end: 38°43′34″.

DMS to decimal degrees

Convert 101°18′24.7″ to decimal degrees, to four decimal places.

101+18/60+24.7/3600=101.3068611…

Rounded to four decimal places: 101.3069°.

Decimal degrees to radians

Convert 137.6° to radians, to three decimal places.

$137.6(\pi/180)=2.401572\ldots$

Keep π in the calculator expression. Round the final answer: 2.402 rad.

Conversion practice

Convert $58.417862°$ to DMS, to the nearest second.

Convert $112°36′18.5″$ to decimal degrees, to four decimal places.

Precision and notation

Minutes and seconds use base 60. A decimal degree is not DMS notation.

Match the question’s requested decimal places. Round the final result once.

Conversion checkpoint

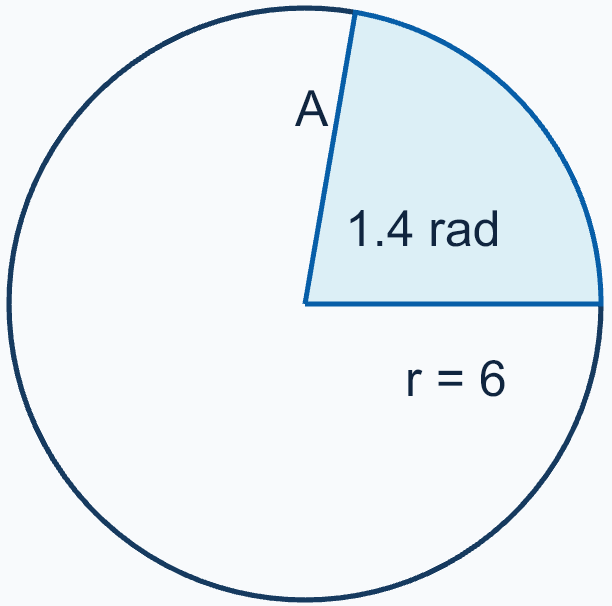
DMS to decimal uses division by 60 and 3600.

Decimal to DMS uses the fractional part twice, multiplying by 60.

Length or area?

The same radius and angle determine both a curved distance and a shaded area.

Which quantity uses square units?



Arc and sector formulas

Arc length: $s=r\theta$. Sector area: $A=\frac{1}{2}r^2\theta$.

Both formulas require θ in radians.

Unknown radius: $r=s/\theta$.

Unknown sector angle: $\theta=2A/r^2$, then convert to degrees if requested.

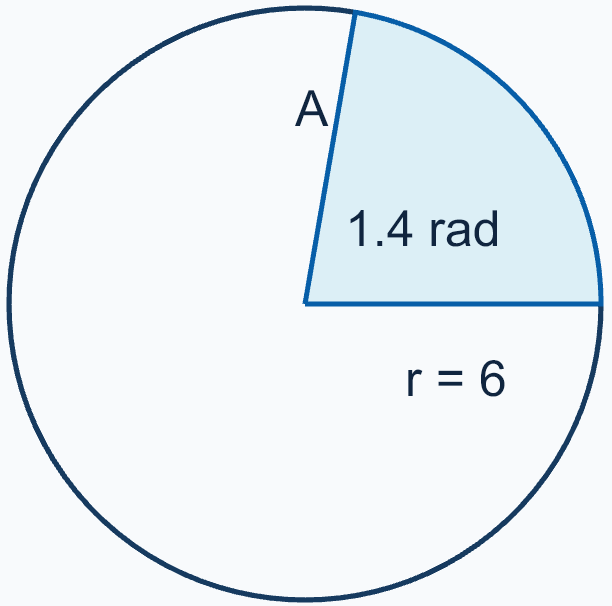
One sector, two measurements

Radius = 6 units and angle = 1.4 rad. Find arc length and sector area.

$s=6(1.4)=8.4$ units.

$A=\frac{1}{2}(6^2)(1.4)=25.2$ square units.

The radius is squared only in the area formula.



Solve backwards for radius

Together: an arc is 24 units long with central angle 80°. Find the radius.

Then independently: an arc is 21 units long with central angle 105°. Find the radius.

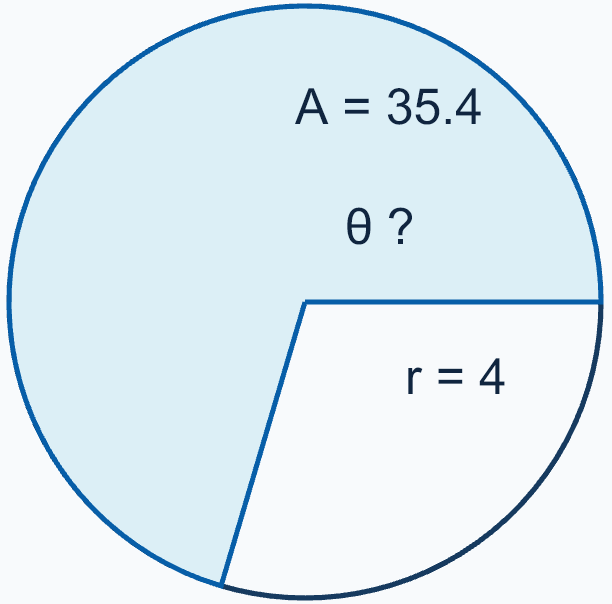
Solve backwards for angle

A sector has area 35.4 square units and radius 4 units. Find θ in radians and degrees.

$\theta = 2A/r^2 = 2(35.4)/4^2 = 4.425$ rad.

$\theta(\text{deg}) = 4.425(180/\pi) \approx 253.53°$.

The angle is between π and 2π radians, so a sector larger than a semicircle is reasonable.

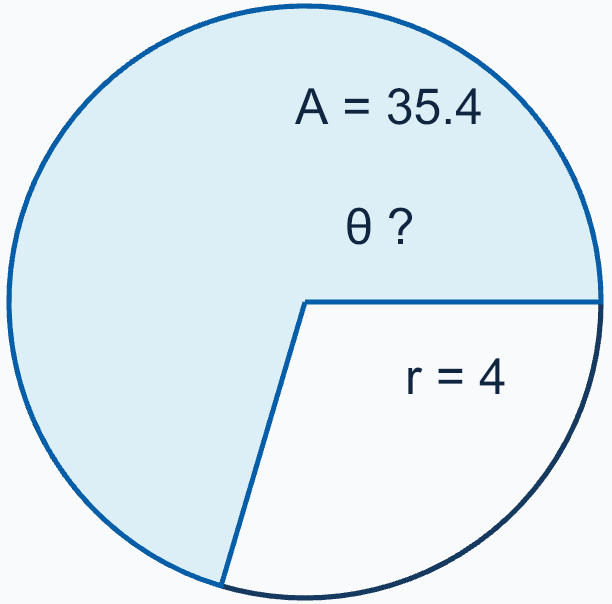


Check an inverse-sector setup

Without looking at the model, write an equation for a sector with area 35.4 and radius 4.

Solve for the angle in radians and degrees.

Explain why an angle above 180° is allowed.



Check the setup before arithmetic

For radius, divide arc length by the angle in radians.

For angle, double the sector area and divide by the squared radius.

A 150° sector with radius 20 has area 500π/3\approx523.60 square units.

Arc and sector checkpoint

Circle the requested unknown first.

Convert degrees to radians before substitution.

Check whether the answer is a length, an area, or an angle.

Same rotation, different distance

Do the center and the rim have the same angular speed?

Does a point farther from the center travel farther per turn?

From rpm to speed

Use radius = diameter ÷ 2.

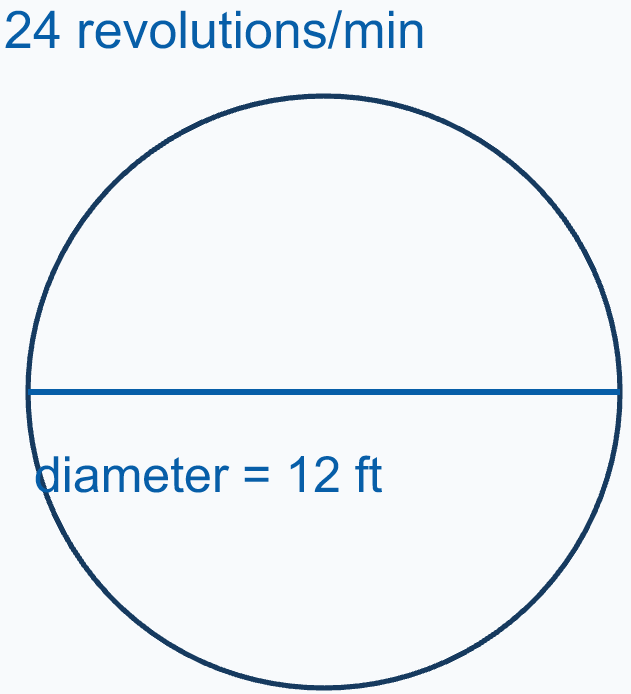
Degrees per second: $n \cdot 360/60$.

Radians per second: $\omega = n \cdot 2\pi/60$.

Linear speed: $v = r\omega$ in distance units per second.

A rotating wheel

Diameter = 12 ft, rotation = 24 rpm. Find angular speed in degrees/s and rim speed in ft/s.



$r=12/2=6$ ft.

$24(360)/60=144°/s$.

$\omega=24(2\pi)/60=4\pi/5$ rad/s.

$v=6(4\pi/5)=24\pi/5\approx15.08$ ft/s.

Rotation practice

A wheel has diameter 8 ft and turns at 30 rpm.

Find angular speed in degrees/s and linear speed at the rim in ft/s.

Write the diameter-to-radius step and both unit conversions.

Two common speed errors

Using the diameter in $v=r\omega$ doubles the speed.

Degrees/s cannot go directly into $v=r\omega$; use radians/s.

Convert minutes to seconds once, and label the final units.

Speed checkpoint

Radius first. Convert rpm. Then multiply radius by radians per second.

A sensible answer includes a distance unit and a time unit.

Independent exit check

An arc is 8π ft long and its central angle is $120°$. Find the radius.

A wheel has diameter 6 ft and turns at 20 rpm. Find angular speed in degrees/s and rim speed in ft/s.